

Do Visual Grounding Decoders Need Feed-Forward Networks?

A Controlled Study over Frozen Vision-Language Features

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<https://github.com/TarunTomar122/attention-is-all-you-need-for-vlms>

Abstract

Do feed-forward networks (FFNs) in visual grounding decoders add essential computation once a pre-trained vision-language model has already encoded image and language context? We compare a four-block attention-only decoder (A4), a matched four-block attention-plus-FFN decoder (S4), and an eight-block attention-only parameter control (A8) over frozen VLM features. A4 matches or slightly exceeds S4 on RefCOCOg and Ref-Adv-s. FineCops-Ref reveals a small A4 deficit of 0.52 percentage points at IoU@0.5 (95% CI [0.12, 0.95] in favor of S4), but A8 recovers it and finishes 0.26 points above S4. Official FineCops levels do not show a monotonic increase in the gap. A4 reduces trainable decoder parameters by 44.4% and cached-decoder latency by 10.1%, although end-to-end latency remains backbone-dominated. These results concern the trainable grounding decoder, not a complete attention-only VLM.

1 Introduction

Visual grounding maps an image and a referring expression to one target box. Modern grounding systems commonly use Transformer decoders that combine self-attention or cross-attention with token-wise feed-forward networks. This design is inherited from the standard Transformer block and remains common in systems that fuse language and image features, including MDETR and Grounding DINO [Kamath et al., 2021, Liu et al., 2024b]. The complete block is a reasonable default when a model must build its representations from scratch. It is less clear whether every sublayer remains necessary in a small task head whose inputs have already been contextualized by a large pretrained model.

This paper isolates that narrower question. A frozen vision-language model supplies spatial image tokens and contextual text states. A small trainable decoder then retrieves the region described by the expression and produces a patch heatmap. We remove only the FFN residuals inside that decoder while holding its frozen features, learned projections, attention layers, readout, loss, optimizer, box conversion, and evaluation protocol fixed. The frozen backbone retains its pretrained FFNs throughout.

The comparison matters because attention and FFNs provide different inductive biases. Cross-attention routes a query toward relevant context. An FFN applies the same token-wise nonlinear map after that routing step. If the frozen features already expose object identity, attributes, spatial location, and language context, the decoder may mostly need retrieval and assembly. If the decoder must construct new relational features, the FFN may remain useful even when attention can localize individual tokens.

We study three questions:

1. At fixed decoder depth, does A4 retain the grounding performance of S4?
2. If a gap appears, does it grow with released or official difficulty metadata?
3. Can additional attention depth recover a gap after the FFN parameter budget is reallocated?

The answer is mixed but coherent. On RefCOCOg direct expressions, A4 is 0.26 percentage points above S4 and satisfies the predeclared retention gate, while the planned direct-versus-logical interaction is not confirmed. On Ref-Adv-s, A4 is 0.79 points above S4 and does not degrade monotonically with negation, expression length, or distractor count. FineCops-Ref exposes the only clear fixed-depth deficit: A4 is 0.52 points below S4. Doubling attention-only depth to A8 recovers that overall gap. These results reject both simple extremes: FFNs are neither universally essential nor uniformly irrelevant in the tested decoder.

Our contributions are:

1. A causal FFN-deletion study over identical frozen image and text features, with a parameter-matched attention-depth control.

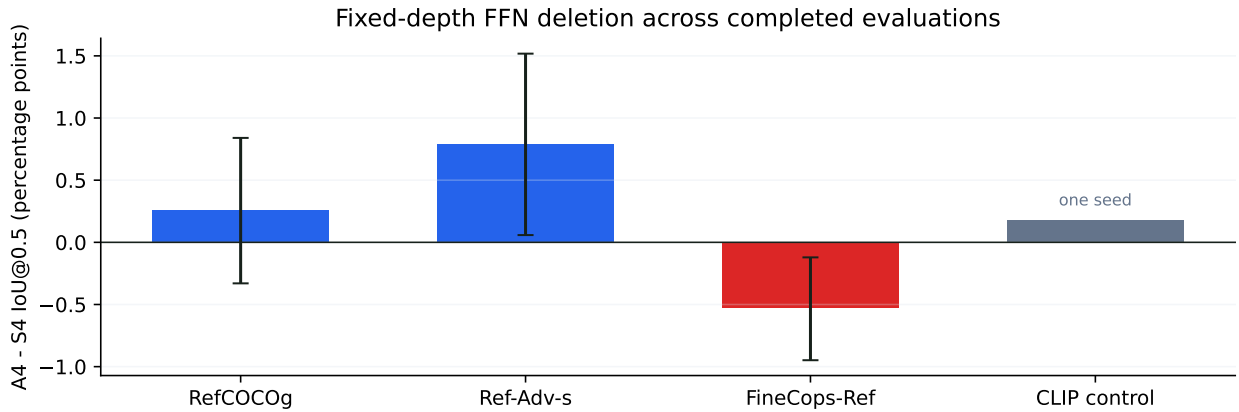


Figure 1: The fixed-depth A4-S4 comparison across completed evaluations. Error bars show the prespecified interval where available; the CLIP control is one-seed descriptive evidence. Positive values favor A4.

2. A difficulty-oriented evaluation across standard, adversarial, and controlled compositional grounding, supported by modality shuffles and paired image-clustered intervals.
3. An evidence-first release that preserves mixed and negative outcomes, including the unconfirmed logical interaction and the absence of a monotonic difficulty boundary.

2 Motivation

2.1 Does a frozen representation change what the decoder needs?

The standard Transformer alternates attention with a position-wise FFN [Vaswani et al., 2017]. Attention moves information between tokens, whereas the FFN transforms each token independently. In a large encoder or generative model, the two operations jointly build and refine representations across many layers. A grounding decoder over frozen VLM tokens operates under a different contract. Its image input already contains a dense raster of contextual features, and its text input already encodes the complete referring expression. The remaining output is one spatial selection.

Prior work establishes that pretrained attention contains useful localization signals. Kang et al. identify a small set of grounding-relevant attention heads inside a frozen LVLM [Kang et al., 2025]. F-LMM translates frozen multimodal attention maps into segmentation masks with CNN and SAM refinement [Wu et al., 2025]. ReCLIP shows that frozen contrastive representations can score candidate regions, although spatial relations remain difficult without an added resolver [Subramanian et al., 2022]. These results make an attention-only decoder plausible, but none isolates whether a newly trained, matched grounding head needs its token-wise FFNs.

Our single-query design sharpens the question. Query self-attention would be degenerate because there is only one query, so each block alternates text and image cross-attention. If the query can retrieve the relevant linguistic and visual features directly, an FFN may add parameters without changing the localization decision. Conversely, compositional expressions may require a nonlinear transformation after retrieval. The experiment therefore needs controlled difficulty evidence rather than one aggregate score.

2.2 Why an overall match is insufficient

A4 could match S4 for several uninteresting reasons. A dataset might contain many direct noun references and too few relational expressions to expose a difference. A learned position prior could produce acceptable boxes without using language. A model could rely on image salience and ignore the expression, or use text length as a shortcut. A heatmap threshold tuned after test access could hide representation differences. Any of these cases would make the architectural conclusion unreliable.

We address these alternatives with a frozen selection boundary and three kinds of controls. First, RefCOCOg includes a predeclared direct-retention test and a direct-minus-logical interaction. Second, image and text shuffles test whether both modalities influence the prediction. Third, Ref-Adv-s and FineCops-Ref provide released metadata for adversarial and compositional difficulty. The study uses only native or official fields for these slices, avoiding a post-hoc taxonomy designed around observed errors.

Uncertainty also matters. Several expressions may describe the same image, so treating all rows as independent would make intervals too narrow. We average paired model seeds per example and bootstrap image IDs, retaining all expressions for every sampled image. A slice is interpreted as a boundary only when its trend and interval support that reading. One noisy level or tuple type does not become a reasoning claim.

2.3 Why capacity reallocation needs a separate control

Removing four FFNs changes both the computation type and the trainable parameter count. At fixed depth, A4 contains 2.64 million trainable parameters while S4 contains 4.74 million. If A4 loses accuracy, that result alone cannot distinguish an FFN-specific advantage from reduced capacity. A matched study therefore needs a model that spends approximately the same parameter budget on attention instead.

A8 repeats the attention-only block eight times and contains 4.75 million trainable parameters, within 0.2% of S4. This control is parameter-matched, not compute-matched: it uses more decoder MACs and higher cached-feature latency. Its purpose is interpretive. If A4 trails S4 and A8 recovers, the evidence supports attention capacity as a substitute at the measured scale. If both A4 and A8 trail S4, the result would be more consistent with an advantage specific to FFN computation.

This distinction follows broader attention-only work. Controlled language-model experiments report that attention-only models can close much of the gap after capacity is reallocated into depth [Ndubaku et al., 2026]. Synthetic grounded composition work also shows that shallow attention can solve attribute composition, with additional depth needed for harder relations [Sikarwar et al., 2022]. Neither result establishes what happens for natural-image grounding over frozen VLM representations, which motivates the present matched decoder study.

3 Methodology

3.1 Frozen features and decoder variants

The primary backbone is frozen SigLIP2 Base/16 at 384-pixel resolution [Tschannen et al., 2025]. For each image-expression pair it returns 576 image tokens arranged in a 24×24 raster and padded contextual text states. Separate learned linear maps project both streams to decoder width $d = 256$. These projections are shared in design and initialization across paired variants and are included in trainable-parameter counts.

The checkpoint is `google/siglip2-base-patch16-384` at pinned revision `f775b65a79762255128c981547af89ad dcf0f88`. The complete image and text encoders run in evaluation mode under `torch.no_grad()`. Final-layer spatial image states remain in raster order, and text padding is preserved as an explicit attention mask. No backbone parameter is fine-tuned, prompt-tuned, LoRA-tuned, or selectively unfrozen. Feature caching is an execution optimization only: cached tensors are required to match uncached tensors before they are used for decoder training or latency isolation.

Images are converted to RGB and resized directly to the backbone’s declared square resolution without cropping. Cropping would be unsafe for grounding because it can remove or truncate the referred object. Ground-truth coordinates are scaled independently along the two axes and the inverse factors are stored for evaluation. The CLIP replication uses the same 24×24 grid at 336-pixel resolution by disabling its default center crop. This keeps decoder token count and output geometry fixed across the two backbone families.

The decoder carries one learned query $q \in \mathbb{R}^{1 \times d}$. Each pre-normalized block first attends to the text tokens T and then the image tokens I :

$$\begin{aligned} q &\leftarrow q + \text{CrossAttentionText}(\text{LN}(q), \text{LN}(T)), \\ q &\leftarrow q + \text{CrossAttentionImage}(\text{LN}(q), \text{LN}(I)). \end{aligned}$$

S4 then applies

$$q \leftarrow q + \text{FFN}(\text{LN}(q)),$$

where the FFN has hidden width 1024 and GELU activation. A4 omits this residual in each of four blocks. A8 uses eight attention-only blocks. Every variant uses eight attention heads, zero dropout, learned affine LayerNorm, residual connections around every retained sublayer, and non-affine per-head query/key RMS normalization.

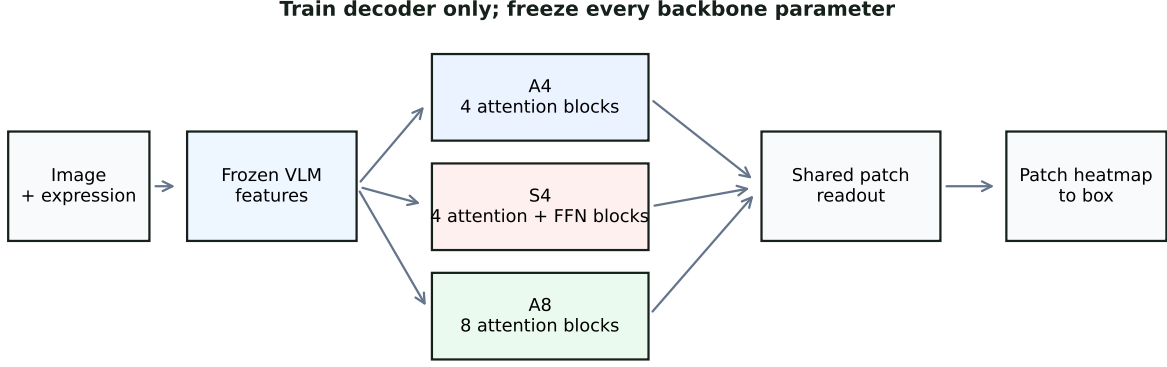


Figure 2: Controlled decoder intervention. The VLM remains frozen and the input projections, patch readout, supervision, and box conversion are shared. A4-S4 isolates FFN deletion; A8 tests parameter reallocation into attention depth.

3.2 Shared patch-attention readout

The prediction head is deliberately separate from the last decoder block. If the model read out the final block’s image-attention weights directly, S4’s last FFN would occur after the prediction and could not affect the loss. The shared readout instead computes a new compatibility score between the final query and every projected image token.

For head h and image patch i ,

$$s_{h,i} = \frac{\langle \text{RMSNorm}(W_h^q q), \text{RMSNorm}(W_h^k I_i) \rangle}{\sqrt{32}}.$$

The readout averages logits across heads and normalizes over all 576 patches:

$$p_i = \text{softmax}_i \left(\frac{1}{H} \sum_{h=1}^H s_{h,i} \right).$$

The distribution p is the model’s only localization output. There is no coordinate-regression MLP, segmentation refiner, CNN, auxiliary language loss, multi-query detector, or backbone adaptation. This keeps the causal path small and prevents an unmeasured nonlinear head from replacing the removed decoder FFNs.

3.3 Target distribution and box conversion

Each ground-truth box is converted to a 24×24 patch target. For patch i , let a_i be the intersection area between the patch cell and the target box. The normalized target is

$$y_i = \frac{a_i}{\sum_{j=1}^{576} a_j},$$

and training minimizes distributional cross-entropy

$$\mathcal{L} = - \sum_{i=1}^{576} y_i \log p_i.$$

At evaluation time, horizontal and vertical marginals are computed from p . For each axis, the decoder selects the shortest contiguous patch interval that contains at least probability mass τ . Ties prefer greater contained mass and then the lower start index. Patch edges are mapped back through the recorded resize and crop geometry to the original image.

The mass threshold was selected once from $\{0.5, 0.6, 0.7, 0.8, 0.9\}$ using S4 on RefCOCOg validation. The selected value, $\tau = 0.8$, was then frozen for every variant, dataset, backbone, and test split. Heatmap metrics remain visible so the conclusion does not depend only on one deterministic box conversion.

3.4 Matched comparisons and controls

The causal fixed-depth comparison is A4 versus S4. Both models receive identical cached frozen features and masks, use the same training budget, and produce the same output shape. A8 is the parameter control. A zero-block decoder (D0), a uniform heatmap, and a fixed position prior provide lower-level diagnostics where available. Image-shuffle and text-shuffle evaluations replace one modality’s pairing while preserving all other code and weights.

The implementation checks that output distributions are finite and sum to one, padding tokens receive zero text-attention probability, ground-truth patch distributions are valid, and predicted boxes remain within image bounds. It also checks that A4 contains no FFN or two-layer token-wise projection and that S4’s final FFN receives gradient from the localization loss. These invariants ensure that the named ablation matches the executed graph.

Paired variants also share ordered example manifests and immutable output directories. Preparation fails on duplicate identifiers, missing images, non-finite coordinates, non-positive boxes, or boxes fully outside an image. Malformed examples are not silently discarded. Raw predictions are stored before aggregate metrics are calculated, which keeps thresholded accuracy, continuous IoU, pointing, target mass, and slice analyses traceable to the same per-example outputs.

3.5 Evaluation and uncertainty

The primary metric is box accuracy at $\text{IoU} \geq 0.5$. Mean IoU, pointing accuracy, and target mass are secondary. Three-seed comparisons average paired seed outputs per example, then use 10,000 paired percentile bootstrap replicates with fixed seed 20260812. Image IDs, rather than expression rows, are resampled so that all references to the same image move together.

For a slice g , define the fixed-depth difference in percentage points as

$$\Delta_g = 100 (A_g^{\text{A4}} - A_g^{\text{S4}}).$$

RefCOCOg direct retention uses its prespecified 90% interval and practical margin. All other reported intervals are 95% descriptive or confirmatory intervals as recorded by the frozen evaluation contract. Ref-Adv-s bins use native metadata quartiles fixed before performance slicing. FineCops uses only official level and tuple-type fields. No LLM-derived semantic categories are introduced after observing results.

4 Experiments

4.1 Models, datasets, and metrics

Backbones. The primary experiments use frozen SigLIP2 features. A one-seed frozen CLIP-L/14@336 experiment checks whether the direction is unique to the primary backbone. This control preserves the decoder definitions, training budget, loss, and frozen mass threshold, but it is descriptive rather than a second multi-seed confirmation.

Data. RefCOCOg UMD is the primary three-seed evaluation and supplies the direct/logical protocol [Yu et al., 2016]. RefCOCO UNC provides a seed-0 classic-dataset replication. Ref-Adv-s contains 1,142 prepared evaluation-only rows and native negation, length, and distractor metadata [Dong et al., 2026]. FineCops-Ref contributes 9,605 official positive-test examples with released difficulty levels and tuple types [Liu et al., 2024a]. No Ref-Adv or FineCops example selects a checkpoint, threshold, architecture, or plot rule.

RefCOCOg contains 80,512 training expressions, 4,896 validation expressions, and 9,602 official UMD test expressions. The classic annotations link each sentence to a COCO object box; multiple expressions may describe one object or image, which motivates image-clustered uncertainty. The RefCOCO seed-0 replication uses its official UNC split and fixed 5,000-update training budget. Ref-Adv-s is the pinned public subset: its packaged split is renamed to test inside this project, and its native caption, box, negation, distractor, image source, and authorship fields are preserved without semantic relabeling.

FineCops evaluation uses only the official positive test examples because the decoder contract predicts one target box for a valid referring expression. Its released level and tuple-type annotations are read directly. All third-party images remain outside Git, and stable IDs plus aggregate or per-example metric tables are released where redistribution permits. This separates reproducible evaluation evidence from upstream image licenses.

Metrics. IoU@0.5 is primary because it captures whether the single predicted box meets the standard localization threshold. Mean IoU retains continuous overlap information, pointing accuracy measures whether the highest-probability patch lies inside the target, and target mass measures the probability assigned to the ground-truth region. Decoder parameters, analytical MACs, peak memory, and repeated synchronized latency measurements describe efficiency.

The primary training and evaluation matrix is intentionally asymmetric. The full three-seed comparison is reserved for the frozen SigLIP2 RefCOCOg models and the locked Ref-Adv-s and FineCops evaluations. RefCOCO and CLIP answer narrow replication questions and are labeled seed-0 or one-seed wherever they appear. This prevents a small descriptive run from being presented with the same evidential weight as the paired multi-seed results.

4.2 Does fixed-depth FFN deletion preserve standard grounding?

On RefCOCOg direct expressions, A4-S4 is +0.26 percentage points with a 90% interval of $[-0.33, +0.84]$. The lower bound remains above the predeclared -5-point practical margin, so direct retention passes. On logical expressions, A4-S4 is -1.04 points with a 95% interval of $[-2.20, +0.15]$. The planned direct-minus-logical interaction is +1.30 points with a 95% interval of $[-0.09, +2.66]$, so the stronger retrieval-versus-reasoning boundary is not confirmed.

The modality controls show that this match does not come from ignoring one input. Correct pairing exceeds image shuffle by 48.7 points for A4 and 48.6 points for S4. Correct pairing exceeds text shuffle by 22.6 and 22.4 points, respectively. Both architectures therefore use image and language information under the tested pipeline. The result supports fixed-depth retention on the primary dataset while preserving the failed interaction as a negative outcome.

The predeclared interaction was deliberately conjunctive: practical retention on direct references and a sufficiently large direct-minus-logical difference were both required. Passing retention alone does not permit the paper to claim that FFNs become necessary for logical language. This gate matters because the point estimate moves in the predicted direction, but its uncertainty still includes zero. Keeping the failed condition visible avoids turning a plausible post-hoc story into a confirmed result.

4.3 Does adversarial difficulty expose an FFN advantage?

Ref-Adv-s is evaluation-only and was accessed after model and threshold choices were frozen. Overall IoU@0.5 is 8.11% for A4, 7.33% for S4, and 7.91% for A8. The paired A4-S4 difference is +0.79 points with a 95% interval of $[+0.06, +1.52]$. The absolute accuracies show that this is a difficult setting for the entire decoder family. The comparison should therefore be read as an architectural stress test, not as a competitive Ref-Adv system.

Negation does not expose a clear A4 loss: the negation slice is -0.15 points with an interval crossing zero, while the non-negation slice favors A4 by 1.41 points. Expression-length quartiles move from +0.34 to +1.79 points without a negative trend. Distractor quartiles range from +0.29 to +1.58 points before ending at 0.00 in the highest-distractor quartile. The longest expressions and largest distractor sets therefore do not produce the hypothesized monotonic attention-only collapse.

This result narrows the interpretation of difficulty. Released metadata can identify useful stress slices, but longer language and more distractors are not equivalent to a controlled requirement for token-wise nonlinear computation. Ref-Adv may also expose limitations in the frozen representation, 24-by-24 readout, or training distribution that affect A4 and S4 together.

The secondary metrics reinforce the bounded reading. A4 and A8 both reach approximately 27% pointing accuracy, and mean IoU differs by less than 0.002 from S4. Thus the positive IoU@0.5 delta is not accompanied by a large shift in continuous overlap. The adversarial set mainly establishes that FFN deletion does not uniquely cause the already-low performance regime; it does not show that any of the three heads solves adversarial reference resolution.

4.4 Does controlled composition reveal a fixed-depth gap?

FineCops-Ref provides the clearest evidence for S4. Across 9,605 official positive-test examples, A4 reaches 27.01% IoU@0.5 and S4 reaches 27.54%. The paired difference is -0.52 percentage points with a 95% interval of $[-0.95, -0.12]$. This is small in absolute size but is the only completed three-seed comparison whose interval lies fully below zero for A4 overall.

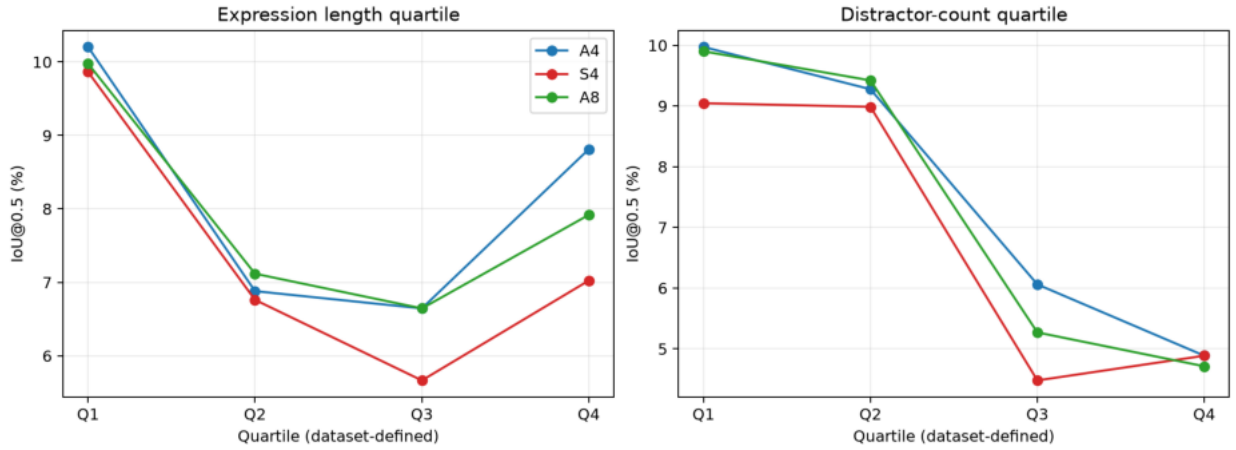


Figure 3: Absolute Ref-Adv-s performance across native difficulty metadata. Absolute values remain low for all variants, so the architectural result is a paired comparison rather than a claim of strong adversarial grounding.

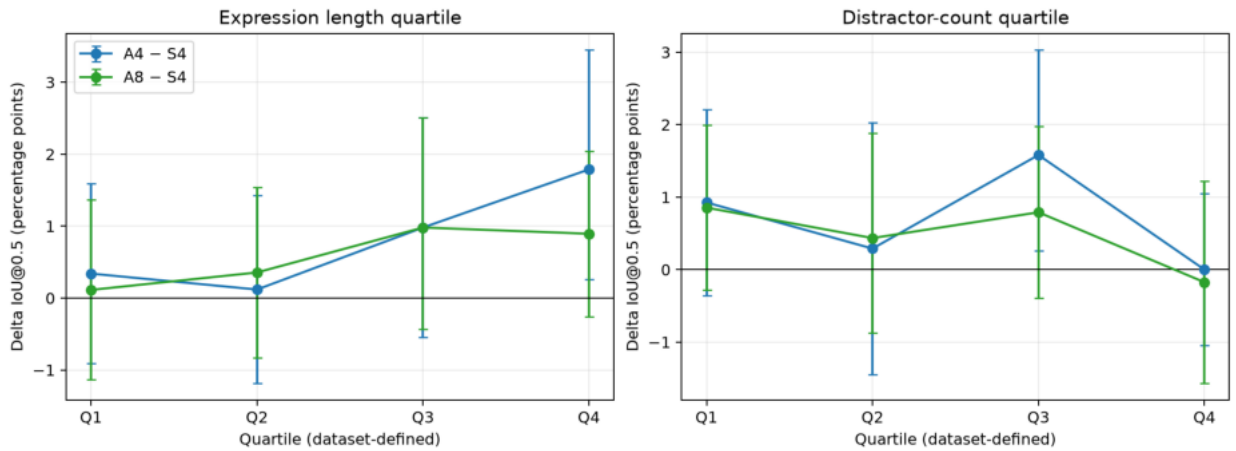


Figure 4: Paired A4-S4 and A8-S4 differences on Ref-Adv-s. Native length and distractor slices do not reveal a monotonic fixed-depth deficit.

The official difficulty levels do not turn that overall deficit into a monotonic boundary. Level 1 is -0.58 points and level 2 is -0.53 points. Level 3 is +0.14 points with a wide interval of $[-1.10, +2.55]$ over 471 examples. The hardest released level therefore does not show a larger A4 loss. It is too imprecise to establish either an A4 advantage or equality.

Continuous metrics show how small the architectural difference remains. Mean IoU is 0.2971 for A4, 0.2978 for S4, and 0.2977 for A8. Pointing accuracy is 52.20%, 51.79%, and 52.04%, respectively, while target mass is 0.3609, 0.3597, and 0.3621. S4’s thresholded advantage therefore does not coincide with a broad advantage on every localization statistic. The paper nevertheless retains the IoU@0.5 result because it is the frozen primary outcome.

Tuple types add a more specific descriptive view. The largest fixed-depth deficit appears in the one-hop slice at -1.32 points with a 95% interval of $[-2.02, -0.49]$. The zero-hop deficit is -0.57 points, while the two-hop slice is -0.08 points with an interval spanning zero. These labels describe the released construction process, not a complete taxonomy of reasoning. Their non-monotonic pattern again argues against the simple rule that more hops imply greater FFN necessity.

4.5 Does attention-only depth recover the gap?

A8 reaches 27.79% on FineCops-Ref, 0.26 points above S4 overall. Its 95% interval relative to S4 is $[-0.34, +0.52]$, so the parameter-matched model is consistent with recovery rather than a new accuracy advantage. At level 2, A8 is +0.70 points above S4; at level 3, it is effectively identical. The two-hop tuple slice favors A8 by 0.99 points with an interval above zero, while the one-hop slice remains -0.23 points with an interval crossing zero.

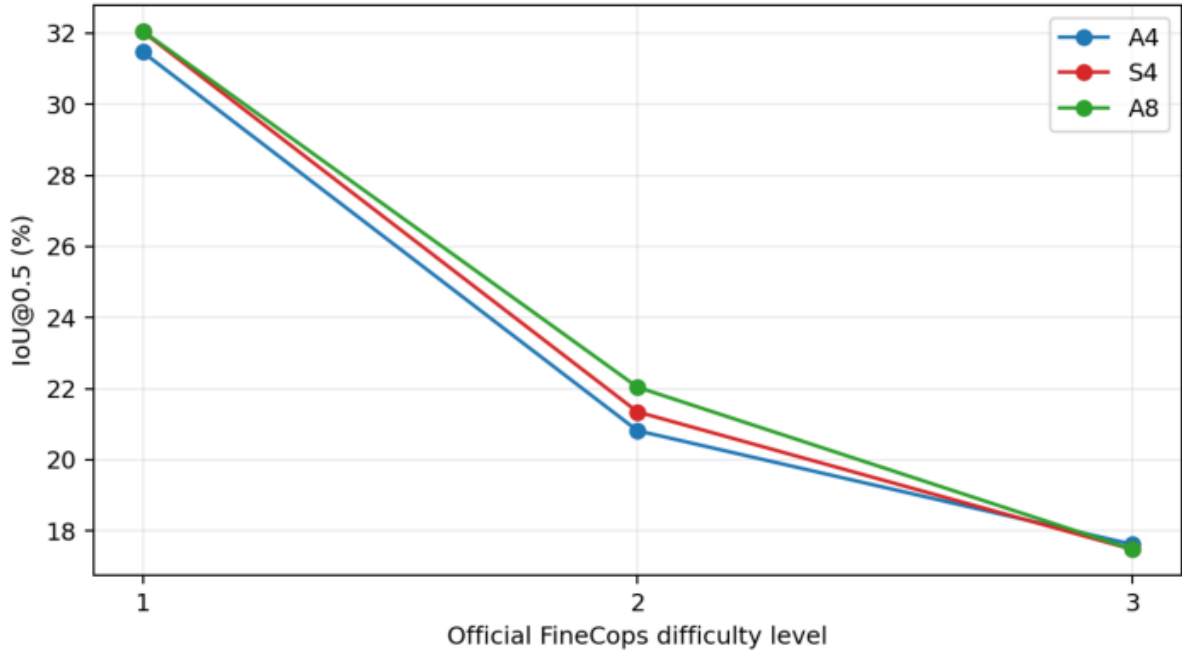


Figure 5: Absolute IoU@0.5 across official FineCops-Ref levels. Performance declines with released difficulty for every decoder; the fixed-depth architectural gap does not grow monotonically.

This pattern changes the causal reading of the A4 deficit. Fixed-depth deletion does remove useful capacity on FineCops, but a comparable parameter budget spent on more attention layers restores the overall score. The result is compatible with attention-only computation performing the needed retrieval and composition when enough depth is available. It does not prove that A8 learns the same internal mechanism as S4.

A8 is also not a compute-matched replacement. Its decoder uses approximately 840 million MACs per example compared with 504 million for S4. The capacity control answers whether attention depth can substitute under a similar trainable parameter count. It does not establish the lowest-latency design.

4.6 Efficiency and second-backbone control

A4 contains 2,639,104 trainable parameters, 44.4% fewer than S4’s 4,743,424. Its analytical MAC count is 501.7 million versus 503.8 million for S4 because cross-attention over the frozen token streams dominates the small FFN difference. Repeated cached-feature timings are 6.71 ms for A4 and 7.46 ms for S4, a 10.1% reduction. Decoder peak memory falls from 817.3 MB to 806.2 MB.

End-to-end timing tells a different story. The complete frozen pipeline takes 209.35 ms for A4 and 210.20 ms for S4, a difference of only 0.4%. Image and text preprocessing plus the frozen SigLIP2 backbone dominate total cost. A4 is therefore smaller and faster as a decoder, but the experiment does not support a meaningful end-to-end speed claim.

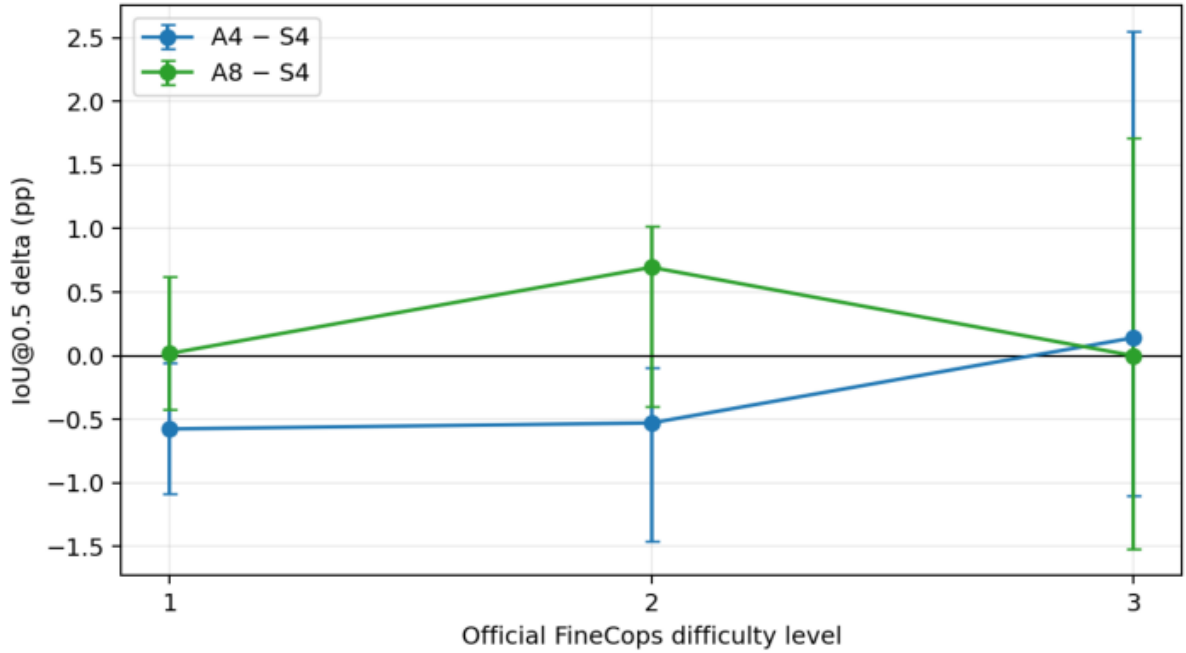


Figure 6: FineCops-Ref paired differences by official level and tuple type. The one-hop slice contains the largest observed A4 deficit, but the sequence is not monotonic.

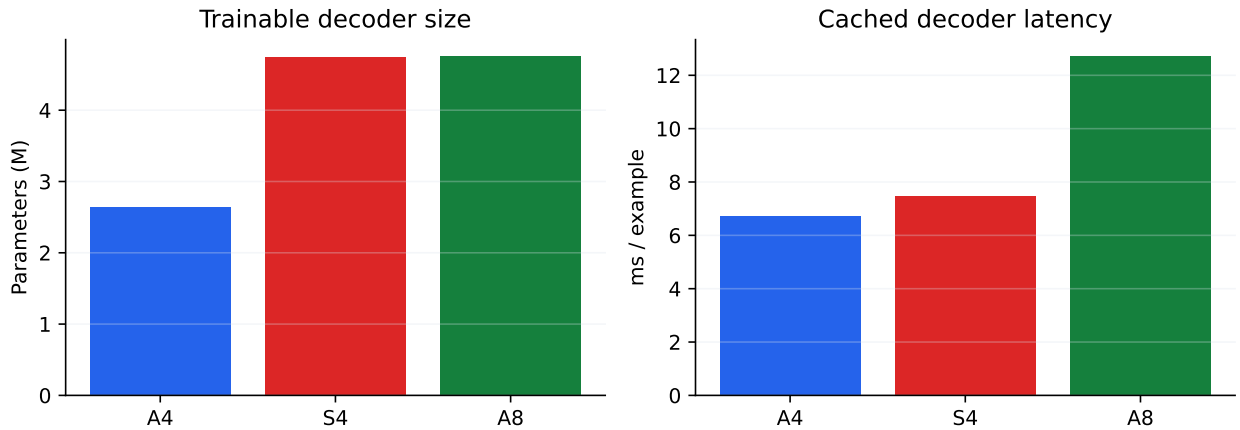


Figure 8: Trainable decoder parameters and cached-feature latency. A4 is the efficient fixed-depth variant; A8 is a parameter control and costs more decoder compute.

The frozen CLIP control checks direction under a second representation family. On 9,602 RefCOCOg examples with one seed, A4 reaches 50.89% and S4 reaches 50.71% IoU@0.5, a +0.18-point difference. Mean IoU differs by -0.22 points, while pointing accuracy is identical. The control is consistent with the primary retention result but remains descriptive because no multi-seed interval was run.

Together, the experiments support four conclusions. First, fixed-depth FFN deletion is benign on standard and adversarial grounding. Second, controlled composition exposes a small but real fixed-depth gap. Third, parameter-matched attention depth recovers that overall gap. Fourth, neither released difficulty levels nor adversarial metadata identifies a monotonic region where FFNs become increasingly necessary.

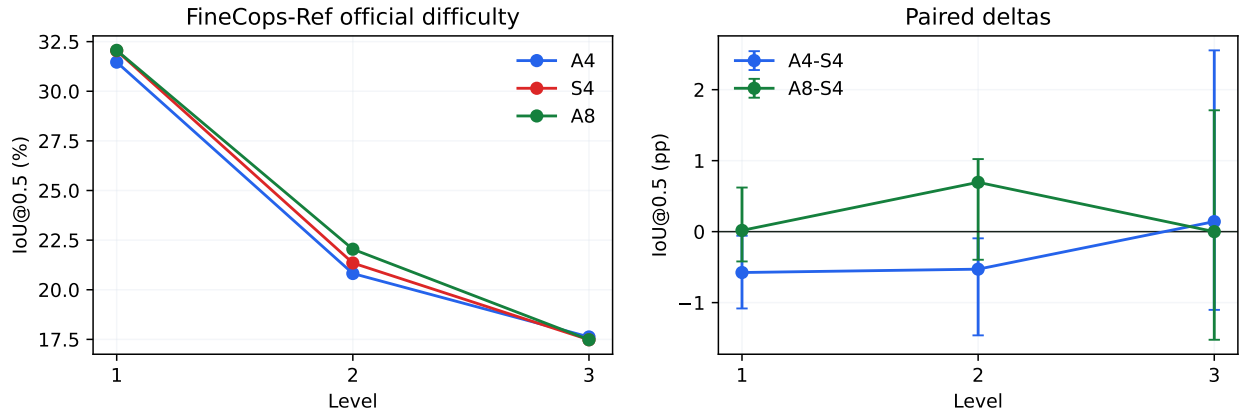


Figure 7: FineCops absolute performance and paired intervals together. A8 recovers the small overall fixed-depth gap, while official level 3 remains too noisy for a difficulty-boundary claim.

5 Limitations

The result applies to one small single-query grounding decoder over frozen features. The backbone itself retains pretrained FFNs and contributes most of the system’s parameters and latency. The paper therefore does not establish a complete attention-only vision-language model, and it does not support removing FFNs from the backbone or a generative language decoder.

The output is one box derived from a 24×24 patch heatmap. The experiment does not cover pixel segmentation, multiple objects, open-vocabulary detection, or autoregressive grounding. Patch resolution and the deterministic mass-to-box rule limit absolute localization quality, particularly on small objects and adversarial examples. A higher-resolution or multi-scale readout could change both absolute scores and the relative value of decoder sublayers.

Only SigLIP2 receives a complete three-seed study. The CLIP experiment is a one-seed directional control, and RefCOCO is seed-0 descriptive evidence. The result therefore samples a narrow set of frozen representations and training runs. No backbone fine-tuning, multiple learned queries, decoder-width sweep, compute-matched attention control, or FFN hidden-width sweep is included.

The FineCops analysis uses the official positive test set and its released metadata. Level 3 and several tuple slices are small or imprecise. Ref-Adv-s is a prepared 1,142-example subset with low absolute accuracy for every variant. These benchmarks are valuable controlled stresses, but their labels should not be treated as exhaustive definitions of compositional or adversarial reasoning.

Finally, the paired bootstrap captures uncertainty over observed images after averaging three seeds. It does not capture uncertainty over backbone pretraining, dataset families, hyperparameter choices, or alternative decoder scales. A8 recovery shows one viable parameter allocation; it does not identify why every FFN effect appears or prove equivalence between the learned computations.

6 Related Work

Transformer grounding decoders. MDETR conditions end-to-end detection on free-form text and jointly reasons over image and language [Kamath et al., 2021]. Grounding DINO combines a feature enhancer, language-guided query selection, and a cross-modality decoder for open-set detection and referring-object benchmarks [Liu et al., 2024b]. VLT uses language-conditioned queries and Transformer interactions for referring segmentation [Ding et al., 2021]. These systems establish attention-plus-FFN decoding as a strong standard design. Our work does not replace their full capabilities; it isolates one FFN ablation in a smaller frozen-feature head.

Grounding from frozen representations. Kang et al. show that a few internal heads in a frozen LVLM carry strong localization signals [Kang et al., 2025]. F-LMM derives word-pixel maps from a frozen multimodal model and retains learned CNN and SAM-based refinement [Wu et al., 2025]. ReCLIP repurposes frozen CLIP for proposal scoring and adds a spatial resolver after showing that off-the-shelf spatial reasoning is weak [Subramanian et al., 2022]. These works demonstrate that pretrained features and attention can support grounding, but they do not compare newly trained decoders with and without FFNs over identical frozen tokens.

Attention-only computation. The original Transformer pairs multi-head attention with position-wise FFNs [Vaswani et al., 2017]. Recent controlled language-model work studies what remains when FFNs are removed and capacity is reassigned to attention depth [Ndubaku et al., 2026]. Sikarwar et al. analyze attention-only composition in synthetic grounded tasks and show that depth matters for relational variants [Sikarwar et al., 2022]. Our experiment asks the corresponding natural-image question under frozen VLM representations and a single-box output.

Referring-expression evaluation. RefCOCO, RefCOCO+, and RefCOCOg are standard sources for referring-expression comprehension [Yu et al., 2016]. Ref-Adv introduces adversarial distractors and reasoning facets designed to suppress common shortcuts [Dong et al., 2026]. FineCops-Ref provides controlled compositional annotations and official difficulty structure [Liu et al., 2024a]. We use these datasets as a sequence of increasingly controlled stress tests, while keeping the model and post-processing frozen after RefCOCOg validation.

7 Conclusion

We asked whether a small visual grounding decoder still needs token-wise FFNs after a pretrained VLM has supplied contextual image and language features. At fixed depth, the answer depends on the evaluation. A4 retains standard RefCOCOg grounding, uses both modalities, and does not collapse on Ref-Adv-s. FineCops-Ref reveals a small 0.52-point deficit, showing that FFN deletion is not free in every controlled setting.

The parameter-matched A8 control recovers the FineCops gap. This shifts the supported conclusion from FFN necessity to capacity allocation: for this single-query decoder, additional attention depth can substitute for the removed FFN budget at the measured scale. Official difficulty levels and adversarial metadata do not reveal a monotonic boundary where attention-only decoding increasingly fails.

The practical result is bounded. A4 is a smaller and lower-latency decoder, but the frozen backbone dominates end-to-end cost. The study does not describe a complete attention-only VLM. It provides controlled evidence that when frozen representations already expose the relevant visual and linguistic context, attention-only decoder capacity can often replace FFN capacity for single-box grounding.

A Reproducibility Details

The release is evidence-first. `scripts/generate_paper_assets.py` reads committed bootstrap, slice, efficiency, CLIP-control, and RefCOCOg summary artifacts. It regenerates the eight manuscript figure families, the main result table, the machine-readable paper-data manifest, and matching web assets without a model download or GPU. `scripts/verify_paper.py` checks the frozen threshold, bootstrap seed and replicate count, headline values, source hashes, expected assets, PDF text, author metadata, and page count.

Artifact	Purpose
<code>configs/study-contract.json</code>	Frozen model, decoder, metric, and bootstrap contract.
<code>research-docs/</code>	Dataset card, final status, and frozen evaluation protocol.
<code>results/</code> and <code>docs/results/</code>	Release map, aggregate evidence, per-example metrics where versioned, bootstrap outputs, and plots.
<code>paper/data/paper-data.json</code>	Machine-readable source snapshot and SHA-256 input hashes for generated paper artifacts.
<code>decision-log/</code>	Compact records of intervention, evaluation, and release decisions.

Table 1: Primary reproducibility artifacts in the release.

The repository intentionally excludes third-party images, backbone weights, provider-local checkpoints, and unversioned qualitative box exports. The paper therefore does not fabricate qualitative examples from aggregate metrics. Qualitative panels should be added only from preserved raw predictions and source-attributed images whose redistribution terms permit publication.

The fixed architecture uses width 256, eight attention heads, four blocks for A4 and S4, eight blocks for A8, FFN hidden width 1024 for S4, zero dropout, and one grounding query. The frozen mass threshold is 0.8. The bootstrap uses 10,000 image-clustered replicates at seed 20260812. `test_study.py` checks architecture, masks, target distributions, gradients, and box-conversion invariants.

The primary backbone revision, dataset preparation decisions, checkpoint selection boundary, and analysis contract are recorded separately from the manuscript. This division is deliberate: the paper presents the scientific argument, while machine-readable contracts and decision logs retain the exact values needed to audit it. Generated figures are derived from versioned JSON, CSV, and Markdown evidence rather than values copied manually into plotting code. Every input file used by the paper-data manifest is stored with its SHA-256 hash.

To reproduce the CPU-only release artifacts, install the pinned Python paper requirements, run `make paper-assets`, and compile `main.tex` with Tectonic. `make verify-paper` repeats generation, compiles the canonical source, reopens the PDF, and checks expected phrases, metadata, page count, figure families, and frozen headline values. Model training and dataset images are not required for this release audit; they are required only to reproduce the upstream experiments themselves.

A.1 Frozen evidence snapshot

Table 2 collects the comparisons that govern the paper’s conclusion. The intervals retain their original roles: RefCOCOg direct uses the prespecified 90% practical-retention interval, Ref-Adv-s and FineCops-Ref use paired 95% intervals, and the CLIP control has no multi-seed interval. The table is generated from the same source artifacts as the figures, so the manuscript does not maintain a second manually edited results ledger.

Table 2: Frozen headline comparisons. Positive A4-S4 values favor the attention-only decoder.

Evaluation	A4-S4 (pp)	Interval	Reading
RefCOCOg direct (3 seeds)	+0.26	90% CI [-0.33, +0.84]	Direct-retention gate passed; interaction not confirmed
Ref-Adv-s overall (3 seeds)	+0.79	95% CI [+0.06, +1.52]	No monotonic hard-slice boundary
FineCops-Ref overall (3 seeds)	-0.52	95% CI [-0.95, -0.12]	A8-S4 = +0.26 pp
CLIP RefCOCOg (1 seed)	+0.18	descriptive only	Second frozen backbone

The decision path is equally important as the point estimates. RefCOCOg passed direct retention but failed the stronger interaction gate. Ref-Adv-s then failed to reveal the expected adversarial boundary. FineCops-Ref supplied the first interval fully favoring S4 at fixed depth, and A8 recovered the overall score under approximate parameter matching. This sequence was preserved rather than rewriting the study around only its final positive result.

Comparison	Question	Permitted reading
A4 versus S4	Does fixed-depth FFN deletion change grounding?	Causal decoder comparison with all other trainable components shared.
A8 versus S4	Can attention depth use a similar parameter budget?	Capacity-reallocation evidence; not compute-matched and not mechanistic equivalence.
Correct versus shuffle	Does the decoder depend on both modalities?	Diagnostic against the simplest image-only or text-only shortcut.
SigLIP2 versus CLIP direction	Does retention depend on one backbone family?	One-seed descriptive transfer only.

Table 3: Interpretation boundaries for the principal controls.

The released evidence is sufficient to regenerate aggregate plots and verify the paper’s numerical claims without proprietary infrastructure. Full training reproduction remains more demanding: it requires the pinned frozen backbones, upstream datasets under their own terms, cached or uncached feature extraction, and the fixed training runs for every seed. Provider-local checkpoints and raw predictions that were not versioned are not silently reconstructed from summary tables. Where those artifacts are absent, the release states the boundary instead of presenting an unverifiable qualitative result.

This audit structure also separates scientific revisions from formatting changes. A later manuscript version may rearrange prose or figures while the machine-readable contract continues to reject altered thresholds, bootstrap settings, headline values, or input hashes. New experiments should enter as new versioned evidence with an explicit decision record; they should not overwrite the frozen first-manuscript snapshot.

A.2 Released difficulty values

Tables 4 and 5 preserve the slice values behind the difficulty figures. They are included to make the non-monotonic conclusion inspectable without estimating values from plots. FineCops levels are official. Ref-Adv quartiles are inclusive empirical bins over released metadata, fixed before decoder-performance differences were examined.

FineCops slice	N	A4	S4	A8	A4-S4 95% CI
Level 1	5,730	31.47	32.04	32.06	[-1.08, -0.06]
Level 2	3,404	20.82	21.35	22.04	[-1.46, -0.09]
Level 3	471	17.62	17.48	17.48	[-1.10, +2.55]
Zero hop	2,333	31.22	31.79	32.30	[-1.33, -0.01]
One hop	2,146	26.72	28.04	27.80	[-2.02, -0.49]
Two hop	2,555	21.97	22.05	23.04	[-0.65, +0.87]

Table 4: FineCops-Ref IoU@0.5 percentages by selected official slices.

The FineCops values show why the overall deficit and the boundary question must remain separate. Levels 1 and 2 favor S4 by similar small amounts, while level 3 reverses sign with much greater uncertainty. The one-hop tuple slice is the largest fixed-depth deficit, but the two-hop slice is near zero and A8 exceeds S4 there. Neither ordering supports a simple more-composition-means-more-FFN rule.

Ref-Adv slice	N	A4	S4	A8	A4-S4 95% CI
Negation	457	7.29	7.44	8.10	[-1.24, +1.02]
No negation	685	8.66	7.25	7.79	[+0.49, +2.34]
Longest quartile	261	8.81	7.02	7.92	[+0.26, +3.45]
Most distractors	191	4.89	4.89	4.71	[-1.05, +1.05]

Table 5: Ref-Adv-s IoU@0.5 percentages on selected native-metadata slices.

Ref-Adv provides the complementary negative result. Negation is inconclusive, the longest-expression quartile favors A4, and the highest-distractor quartile is tied at fixed depth. These slices do not identify a region where S4’s FFNs become increasingly advantageous. Their low absolute scores remain visible in the table and prevent a paired retention result from being mistaken for strong adversarial performance.

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References

- Henghui Ding, Chang Liu, Suchen Wang, and Xudong Jiang. Vision-language transformer and query generation for referring segmentation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 16321–16330, 2021.
- Qihua Dong, Kuo Yang, Lin Ju, Handong Zhao, Yitian Zhang, Yizhou Wang, Huimin Zeng, Jianglin Lu, and Yun Fu. Ref-adv: Exploring mllm visual reasoning in referring expression tasks, 2026.
- Aishwarya Kamath, Mannat Singh, Yann LeCun, Gabriel Synnaeve, Ishan Misra, and Nicolas Carion. MDETR: Modulated detection for end-to-end multi-modal understanding. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 1780–1790, 2021.
- Seil Kang, Jinyeong Kim, Junhyeok Kim, and Seong Jae Hwang. Your large vision-language model only needs a few attention heads for visual grounding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9339–9350, 2025.
- Junzhuo Liu, Xuzheng Yang, Weiwei Li, and Peng Wang. Finecops-ref: A new dataset and task for fine-grained compositional referring expression comprehension, 2024a.
- Shilong Liu, Zhaoyang Zeng, Tianhe Ren, Feng Li, Hao Zhang, Jie Yang, Qing Jiang, Chunyuan Li, Jianwei Yang, Hang Su, Jun Zhu, and Lei Zhang. Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection. In *European Conference on Computer Vision*, 2024b.
- Henry Ndubuaku, Karen Mosoyan, Jakub Mroz, Noah Cylich, Satyajit Kumar, Parkirat Sandhu, Roman Shemet, and Justin H. Lee. A controlled study of attention-only transformers, 2026. URL <https://arxiv.org/abs/2607.18363>.
- Ankur Sikarwar, Arkil Patel, and Navin Goyal. When can transformers ground and compose: Insights from compositional generalization benchmarks. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 648–669, 2022.
- Sanjay Subramanian, William Merrill, Trevor Darrell, Matt Gardner, Sameer Singh, and Anna Rohrbach. ReCLIP: A strong zero-shot baseline for referring expression comprehension. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*, pages 5198–5215, 2022.
- Michael Tschannen, Alexey Gritsenko, Xiao Wang, Muhammad Ferjad Naeem, Ibrahim Alabdulmohsin, Nikhil Parthasarathy, Talfan Evans, Lucas Beyer, Ye Xia, Basil Mustafa, Olivier Henaff, Jeremiah Harmsen, Andreas Steiner, and Xiaohua Zhai. Siglip 2: Multilingual vision-language encoders with improved semantic understanding, localization, and dense features, 2025.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- Size Wu, Sheng Jin, Wenwei Zhang, Lumin Xu, Wentao Liu, Wei Li, and Chen Change Loy. F-lmm: Grounding frozen large multimodal models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 24710–24721, 2025.
- Licheng Yu, Patrick Poirson, Shan Yang, Alexander C. Berg, and Tamara L. Berg. Modeling context in referring expressions. In *European Conference on Computer Vision*, pages 69–85, 2016.